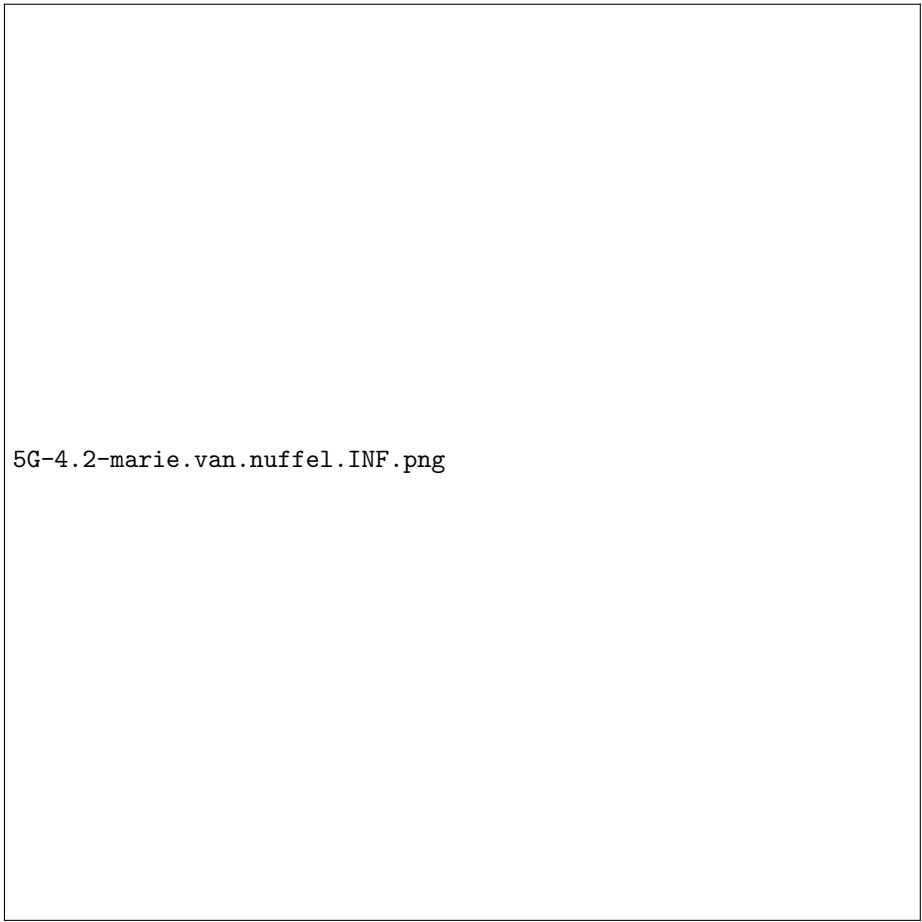


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Oefeningenreeks 5G nr. 4.2

$f(y) = y(3 - y)$ op het interval $[1,3]$

$$\begin{aligned} & \pi \int_1^3 (3y - y^2)^2 dy \\ & \pi \int_1^3 (9y^2 - 6y^3 + y^4) dy \\ & = \pi \left[9 \cdot \frac{y^3}{3} - 6 \cdot \frac{y^4}{4} + \frac{y^5}{5} \right]_1^3 \\ & = \pi \left[\left(9 \cdot \frac{27}{3} - 6 \cdot \frac{81}{4} + \frac{243}{5} \right) - \left(9 \cdot \frac{1}{3} - 6 \cdot \frac{1}{4} + \frac{1}{5} \right) \right] \\ & = \pi \left[3 \cdot 27 - 81 \cdot \frac{3}{2} + \frac{243}{5} - 3 + \frac{3}{2} - \frac{1}{5} \right] \\ & = \pi \left[\frac{810}{10} - \frac{1215}{10} + \frac{486}{10} - \frac{30}{10} + \frac{15}{10} - \frac{2}{10} \right] \\ & = \frac{32\pi}{5} \end{aligned}$$

References



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